

Meghna Pawle

meghnapawle27@gmail.com
+91 9082368339
LinkedIn

GitHub
CodeChef
Codeforces
LeetCode

Profile Summary

Pre-Final year B.Tech student in Artificial Intelligence at NIT Surat with strong fundamentals in Data Structures, Algorithms, and Object-Oriented Programming. Proficient in C++ and Python, with hands-on experience building scalable software systems and machine learning pipelines.

Education

- **Sardar Vallabhbhai National Institute of Technology** Surat, India
B.Tech in Artificial Intelligence Aug 2024 – Dec 2028
 - Junior Developer, Google Developer Group on Campus
 - Executive, Association of Computing Machinery
 - Executive, Cutting Edge Visionaries
 - Coordinator, Mindbend — Gujarat's Largest Techno-Managerial Fest
- **St. Mary's Junior College** Navi Mumbai, India
Higher Secondary Education Aug 2021 – Jul 2023

Projects

- **Multithreaded Autocomplete Engine**
 - *Tools: C++, STL, Multithreading, Synchronization, Git*
 - Engineered a thread-safe **Trie-based** autocomplete system achieving **0.09 ms average prefix search latency** under concurrent read/write workload.
 - Processed **130K+ words at 275K words/sec** using a producer-consumer pipeline with bounded queue (10,000), enabling realtime frequency-ranked autocomplete during active ingestion.
 - Ensured stability and scalability with atomic metrics, clean shutdown, and **deadlock-free** thread coordination.
- **Mutual Fund Return Prediction Model**
 - * *Tools: Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Google Colab*
 - Developed **Random Forest-based classification models** to predict top-performing mutual funds across **1Y, 3Y, and 5Y horizons** using 28 financial and risk features.
 - Achieved up to **94% ROC-AUC** and **91% accuracy**, enabling reliable long-term fund performance predictions.
 - Processed and analyzed data from **647 mutual funds**, performing feature engineering, model evaluation, and automated probability-based ranking
- **BlueMesh — Bluetooth Mesh Messenger App**
 - *Tools: Android Studio, Bluetooth Low Energy (BLE), Kotlin*
 - Built a BLE-based messaging app enabling secure one-to-one device communication with discovery and message transfer.
 - Integrated foundational mesh networking protocols for potential multi-device communication (currently in development).
 - Designed a clean and intuitive UI to enhance usability and interaction flow.

Highlights

- * Team Leader — Finalist, Web Wonders Hackathon — Nexus NIT Surat
- * Active participant in Codeforces and CodeChef contests (Rating: 1260); solved 100+ Data Structures and Algorithms problems across platforms like LeetCode and through extensive offline C++ practice.

Skills

- * **Languages:** C/C++, Python, JavaScript, SQL
- * **Tools:** Linux, HTML, CSS, Git/GitHub, VS Code, Jupyter Notebook, Android Studio, Google Colab
- * **Machine Learning:** NumPy, Pandas, Matplotlib, Scikit-Learn, Linear & Logistic Regression, Decision Trees, Random Forest